

RecoverX: Fraud-Aware Onboarding Engine

Technical Documentation & System Architecture

1. Core Mission

RecoverX is a production-grade AML intelligence layer designed to detect mule accounts at the point of creation. It provides real-time risk scoring (TIS) across 5 intelligence modules.

2. Key File Descriptions

- `src/services/onboardingEngine.js`: The brain. Handles Identity, Device, Telecom, Velocity, and Behavioral checks.
- `src/pages/Onboarding.jsx`: The UI wizard. Manages the 6-phase onboarding flow and decision logic.
- `src/services/mockData.js`: Pattern matching backend. Detects ID/Name reuse from existing data.
- `vercel.json`: Permanent routing configuration for live deployments.

3. The 5 Intelligence Modules

- A. Identity: PAN/Aadhaar validation and multi-account reuse detection.
- B. Device: Fingerprinting, VPN/Proxy detection, and Emulator flagging.
- C. Telecom: Disposable email/phone domain blacklist and newly-activated SIM detection.
- D. Velocity: Signup frequency monitoring per IP and Device ID.
- E. Behavioral: Silent telemetry of keystrokes, paste events, and form completion speed.

4. Risk Scoring (TIS)

Scores are weighted from 0-100. Over 60 triggers a 'HIGH RISK' block, forwarding the case to the investigator queue with a full audit trail.